

Musa Rioba

Data Engineer | Author | Technical Trainer

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience building production-grade data systems, ML pipelines, and interactive analytics platforms. Published author of Production Data Science with R. Actuarial Science background with deep expertise in ETL/ELT orchestration, real-time systems, risk modeling, and compliant data infrastructure. 4 years of technical training experience teaching data analysis and programming to 100+ professionals.

TECHNICAL SKILLS

Languages: R (Expert), Python, SQL, Bash, HTML, CSS, Flutter, Dart

Data Engineering: Apache Airflow, dbt, PySpark, ETL Pipelines, ELT Pipelines, Data Modeling, Web Scraping, API Integration

Machine Learning: tidymodels, XGBoost, Prophet, K-Means, RFM Analysis, Markov Chains, Isolation Forests, Risk Scoring

Visualization: R Shiny, ggplot2, Plotly, shinydashboard, DT DataTables

Databases: PostgreSQL, Snowflake, SQLite, Redis, BigQuery

DevOps: Docker, Docker Compose, GitHub Actions, Git, GitHub, pre-commit, Nginx, Plumber APIs, Fly.io

Statistics: A/B Testing, Sequential Testing, Bayesian Methods, Statistical Modeling, Actuarial Methods

PROFESSIONAL EXPERIENCE

Freelance Data Engineer and Consultant — Self-Employed

2019 – Present

- Architected production ETL platforms ingesting multi-source sales, customer, and product data using Apache Airflow, dbt, and PySpark with full data quality gates and CI/CD pipelines.
- Built a real-time fraud detection system using Redis and isolation forests, processing high-velocity transactions.
- Developed demand forecasting ensembles (Prophet plus XGBoost) and Bayesian A/B testing frameworks for marketing analytics.
- Deployed interactive dashboards (R Shiny, Plotly) for clinical trial monitoring, sales analytics, and insurance marketplace platforms.

Technical Trainer — Various Institutions

2019 – 2023

- Designed and delivered data analysis, R, and Python curricula to 100+ students and professionals.
- Taught production best practices: reproducibility, scalability, observability, and safety in data pipelines.
- Developed hands-on workshops on ETL orchestration, statistical modeling, and dashboard development.

SELECTED PROJECTS

Telematics Kenya — UBI Risk Platform (Flutter, R, Docker, Fly.io)

- Full-stack usage-based insurance platform for the Kenyan motor insurance market with a Flutter mobile app (Telematics Kenya) and a web-based backend admin dashboard deployed on Fly.io.
- Flutter mobile app features real-time trip tracking, risk score gauge, fleet overview KPIs, and driver profile management.
- Web backend dashboard provides driver profiles, risk scoring, trip history with interactive maps, and model performance monitoring.
- Weighted ensemble risk model (Gini 0.41, AUC-ROC 0.73) with SHAP-like explanations and fairness audits across gender, age, and region.
- R Plumber REST API with 10+ endpoints; Docker Compose stack (Nginx, R API, PostgreSQL, Redis) with GDPR compliance controls.

Production ETL/ELT Data Platform (Python, Airflow, dbt)

- End-to-end data platform with 3 Airflow DAGs, dbt medallion architecture, and dual-mode processing using Pandas and PySpark.

- Multi-source ingestion (CSV, Snowflake, REST API) with runtime validators, dbt tests, and structured JSON logging.
- Docker Compose stack tuned for 16GB RAM with GitHub Actions CI/CD, pre-commit hooks, and pytest test matrix.

Clinical Trial Monitoring Dashboard (R Shiny, Docker)

- Phase II diabetes trial system with CDISC SDTM compliance, role-based access control across 3 roles, adverse event and serious adverse event tracking, and audit logging.
- Deployed via Docker and Shinyapps.io with GitHub Actions CI/CD pipeline for linting and data integrity tests.

Kenya Insurance Marketplace Platform (R Shiny)

- Interactive dashboard with 4 premium calculators (Motor, Health, Life, Agriculture) across 28 licensed insurers.
- 8 structured datasets, regulatory compliance tracking (IRA), NHIF benefits, and market analysis visualizations.

Open Source R Packages

- `ugcensus`: Toolkit for Uganda census data 1991 to 2024 with multi-level geography and population projections.
- `sacensus`: Toolkit for South African census data 1996 to 2022 with demographic analysis and trend visualization.
- `kencensus`: Toolkit for Kenya census data 1948 to 2019 covering 13 themed datasets across 47 counties.

EDUCATION

BSc Actuarial Science — Karatina University, Kenya

PUBLICATIONS

Production Data Science with R — 8-chapter comprehensive guide to production-grade ML pipelines, ETL, fraud detection, and A/B testing.

ADDITIONAL

- 4 years technical training experience; 100+ students taught
- Open source contributor
- Docker, GitHub Actions, CI/CD